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8 December 1986
SUPERSEDING
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MILITARY SPECIFICATION

BLENDER, POWDERED MIX, MECHANICALLY REFRIGERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanically refrigerated (sea water cooled) powdered mix blenders.

1.2 Classification. The blenders covered by this specification shall be of the following types as specified (see 6.2).

Type I: Floor mounted
Type II: Table top mounted

1.2.1 Part numbers. The part numbers for the blenders covered by this specification shall be as follows:

a. Type I: M44037-01
b. Type II: M44037-02

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.
FF-C-77 - Casters, Rigid and Swivel (Institutional Duty).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-741 - Steel, Carbon: Structural Shapes; Plates; and Bars.

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic and Associated Electrical Equipment).
MIL-F-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing; General Specification for.
MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing of.
MIL-R-14328 - Rubber Sheet: Synthetic, Medium Soft, General-Purpose, Gasket Material (For Extreme Climatic Conditions).
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-B-16541 - Bronze, Valve: Castings.
MIL-F-20042 - Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-B-24059 - Bronze, Nickel Aluminum; Rod, Flat Products with Finished Edges, Shapes and Forgings.
MIL-I-43717 - Ice Milk, Milk Shake and Yogurt Mixes.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip, Specification for.
A276 - Stainless and Heat-Resisting Steel Bars and Shapes, Specification for.
A582 - Free Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished, Specification for.
B61 - Steam or Valve Bronze Castings, Specification for.
B148 - Aluminum Bronze Sand Castings, Specification for.
B150 - Aluminum Bronze Rod, Bar, and Shapes, Specification for.
B280 - Seamless Copper Tube for Air Conditioning and Refrigeration Field Service, Specification for.
B584 - Copper Alloy Sand Castings for General Applications, Specification for.
C177 - Steady-State Thermal Transmission Properties by Means of the Guarded Hot Plate, Test Method for.
D1562 - Cellulose Propionate Molding and Extrusion Compounds, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

DEPARTMENT OF HEALTH AND HUMAN SERVICES (DHHS)

Federal Food, Drug, and Cosmetic Act, Food Additive Amendment for Contact with Food.

(Application for copies should be adressed to the Department of Health and Human Services, Food and Drug Administration, 330 Independence Avenue, S.W., Washington, DC 20204.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
ICS 2 - Standards for Industrial Control Devices, Controllers, and Assemblies.
ICS 6 - Enclosures for Industrial Controls and Systems.
MG 1 - Motors and Generators.
WC 5 - Thermoplastic-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

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WC 7 - Cross-Linked-Thermosetting-Polyethylene-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.
WD 1 - General-Purpose Wiring Devices.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Battermarch Park, Quincy, MA 02269.)

NATIONAL SANITATION FOUNDATION (NSF)

No. C-2 - Special Equipment and Devices.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 498 - Electrical Attachment Plugs and Receptacles.
UL 621 - Ice Cream Freezers and Soda Fountain Units.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. The blenders shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the blenders being furnished. A standard commercial product is a product

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which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Codes and standards. The blenders shall conform with the applicable requirements of NSF C-2, UL 621, and the DHHS Federal Food, Drug, and Cosmetic Act, Food Additive Amendment for Contact with Food.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4.1 Refrigeration tube. Copper tube refrigeration lines shall conform to ASTM B280. Tube used for the fabrication of the evaporator and condenser shall be the type normally used for the purpose by the manufacturer except it shall have a wall thickness of not less than 0.016 inch [0.4 millimeter (mm)].

3.4.2 Bolts, nuts, screws, and washers. Bolts, nuts, screws, and washers used for mounting components shall be of the quality and in the sizes, types, and materials normally used for the purpose in commercial practice. Threads shall conform to FED-STD-H28/2. Steel bolts, nuts, screws, and washers for the evaporator shall be zinc plated. Brass bolts, nuts, washers, and machine screws shall not be used in fastening aluminum parts.

3.4.3 Insulation. All insulation shall be expanded polyurethane and shall have a "K" factor, based on ASTM C177 determined by using a 75 degrees Fahrenheit (°F) [25° Centigrade (C)] mean temperature, expressed in British thermal units per hour per square foot per inch of thickness per °F temperature difference. Loose fill type insulation shall not be acceptable.

3.4.3.1 Foamed-in-place. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial "K" factor of not greater than 0.15 and a density of not less than 1.9 pounds per cubic foot (30.43 kilograms per cubic meter) when tested in accordance with 4.7.

3.4.4 Plastic. The plastic shall be cellulose propionate conforming to type I, class 3, grade 1 of ASTM D1562.

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3.4.5 Stainless steel plates, sheets, and strips. The stainless steel plates, sheets, and strips shall conform to any type 300 series of ASTM A167.

3.4.6 Stainless and heat-resisting steel bars. Stainless and heat-resisting steel bars shall conform to ASTM A276 or ASTM A582.

3.4.7 Structural steel. Structural steel shall conform to QQ-S-741.

3.5 Design. The blender shall be a self-contained unit, designed to thoroughly blend water and dehydrated powdered mix conforming to type III of MIL-I-43717. The type I blender shall be designed to have a production capacity of not less than 20 quarts [18.9 liters (L)] of mixture and the type II blender shall be designed to have a production capacity of not less than 10 quarts (9.5 L) of mixture. The blenders shall be designed to cool the mixture to 40°F (4°C). The blenders shall consist of a welded structural steel frame and base, exterior panel assembly, mix system, electrical system, refrigeration system, manual water fill control, and controls necessary for automatic operation.

3.5.1 Electrical characteristics. Unless otherwise specified (see 6.2), the blender shall be designed for operation on 440 volts, 60 Hertz, 3-phase system. Wiring shall be enclosed, supported, and protected from grease or food spillage and against mechanical damage. Metal edges with which wire comes in contact shall be rounded and smooth. Wires shall be secured to prevent interference with servicing and operation. Wiring shall be in accordance with NEMA WC 5 or NEMA WC 7. Wiring practices shall be in accordance with NFPA 70.

3.5.2 Repair and maintenance. The blenders shall be designed to allow ready adjustment, service, and replacement of electrical power system components and operational electrical accessories and mechanical components, without disturbing or disassembly of any major component as required, for proper operation of the blenders.

3.6 Construction. The blenders shall be provided with 0.50 inch iron pipe size (ips) inlet and outlet water connections located at the back of the unit. All accessories, instruments, controls, and tubing shall be fastened or braced. All refrigerant components shall be gas tight when tested in accordance with 4.6.3. Contact areas between dissimilar metals, except freezer head bearings, shall be insulated to minimize galvanic corrosion by coating the surface of both metals with zinc chromate or asphalt type compounds or by strips of rubber conforming to MIL-R-14328. Dissimilar metals are specified in MIL-STD-889. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without removing any other major component. The blenders shall be provided with four legs suitable for bolting the blender to the deck or table top as applicable. The bottom flange of each leg shall be provided with clearance holes for mounting bolts. When specified (see 6.2), the type I blender shall be furnished with 3-inch (76 mm) diameter casters conforming to type II, class C, style C mounting of FF-C-77. The casters shall be equipped with brakes. Unless

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otherwise specified (see 6.2), the dimensions of the the type I blender shall be not greater than 16 inches [40.6 centimeters (cm)] in width, 29 inches (73.7 cm) in depth, and 59 inches (150 cm) in height including legs, with a dimensional tolerance of ±1 inch (±25 mm) for all dimensions. Unless otherwise specified (see 6.2), the dimensions of the type II blender shall be not greater than 16.25 inches (41.3 cm) in width, 33 inches (83.8 cm) in depth, and 37 inches (93.4 cm) in height including legs.

3.6.1 Exterior panel assembly. The exterior panel assembly shall consist of side panels (left and right) back and front panels and cabinet top. Components of exterior panel assembly shall be fabricated from 0.050 inch (1.27 mm) thick stainless steel in accordance with 3.4.5. Each side panel shall be provided with louvers. The front panel shall incorporate the mixing chamber door, manual water fill control, indicator light(s) and removable container shelf. The container shelf shall be fabricated of 0.062 inch (1.6 mm) thick stainless steel in accordance with 3.4.5. The cabinet top shall be provided with a filler hole not less than 2 inches (51 mm) in diameter with cover. The lip of the filler hole shall be raised above the cabinet top. The lip of the filler hole cover shall be hinged and fabricated of stainless steel in accordance with 3.4.6. The exterior panels and components shall be easily removable by the use of simple tools.

3.6.2 Mix system. The mix system shall consist of a mixing chamber, mixing chamber door assembly, beater assembly, cross-bar assembly, drip pan and other components.

3.6.2.1 Mixing chamber. The mixing chamber shall be fabricated from 0.078 inch (2.0 mm) thick stainless steel in accordance with 3.4.5. The complete exterior area of the mixing chamber shall be covered with insulation in accordance with 3.4.3. The completely assembled mixing chamber shall have a capacity of 20 quarts (18.9 L) for the type I and 10 quarts (9.5 L) for the type II.

3.6.2.2 Mixing chamber door assembly. The mixing chamber door assembly shall consist of a clear plastic door, draw valve unit, cross-bar, and cross-bar adjusting screw and door gasket. The clear plastic door shall be fabricated from nominal 0.75 inch (19 mm) clear plastic in accordance with 3.4.4. The face of the plastic door shall be provided with a permanent fill indicator line and shall incorporate the draw valve unit. The rear of the plastic door shall be grooved to accept the door gasket. The draw valve unit shall be fabricated from plastic in accordance with 3.4.4 and shall contain an outer housing, piston, "O" ring and draw valve handle. The draw valve handle, cross-bar, and cross-bar adjusting screw shall be fabricated from type 302 or 304 stainless steel in accordance with 3.4.6.

3.6.3 Refrigeration system. The refrigeration system shall be self-contained within the framework. The refrigeration system shall consist of an electric motor driven refrigeration compressor, condenser, controls, and all other accessories, necessary to operate as a completely automatic refrigeration system.

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3.6.3.1 Compressor. The compressor shall be designed to meet the requirements of 3.5 using refrigerant conforming to type 12 of BB-F-1421. The compressor shall be of the semi-hermetic or fully hermetic type with electric motor and compressor components in a common housing. The suction and discharge service valves shall have provisions for gage attachments and shall be bolted to the compressor and shall be easily removable for compressor replacement. The compressor shall be installed on not less than two of the four resilient external mountings. A disposable type dehydrator shall be provided.

3.6.3.2 Dehydration and charging. The completely assembled refrigeration system shall be evacuated, dehydrated, and charged with the refrigerant conforming to type 12 of BB-F-1421 and the quantity of oil recommended by the compressor manufacturer.

3.6.3.3 Moisture and liquid indicator. A moisture and liquid indicator shall be installed in the liquid line. It shall indicate dry when the refrigeration system is tested in accordance with 4.6.4.

3.6.3.4 Sea water condenser. The condenser shall be designed for use with sea water at an entering temperature between 85°F (29°C) and 110°F (43°C) condensing temperature. Water velocities in the condenser shall be not greater than 6 feet [1.8 meters] per second at design conditions. The condenser shall be capable of being mechanically cleaned and installed so that the tubes can be cleaned without removing the condenser from the equipment. All parts in contact with sea water shall be made from copper-nickel composition 90-10 conforming to MIL-C-15726, aluminum bronze conforming to MIL-B-24059, copper alloy 903 conforming to ASTM B584, aluminum bronze 955 conforming to ASTM B148, copper alloy 630 conforming to ASTM B150, or valve bronze conforming to ASTM B61. Condenser sea water tubes shall be fabricated from copper-nickel composition 90-10 conforming to MIL-C-15726. The heads may be made of Delrin or a similar material satisfactory to Naval Sea Systems Command. The condenser shall be provided with devices to purge both the refrigerant and water circuits and for draining the water circuit. A water regulating valve shall be provided with each condenser. The valve shall be of the direct-acting or pilot-controlled type actuated by condenser gas pressure to modulate flow of water required for the condenser. The valve shall regulate the water flow from shut-off to required capacity within an operating gas pressure rise of not greater than 40 pound-force per square inch (psi) [275,790 Pascals (Pa)]. The valve shall be adjustable within an operating range of 90 to 140 psi (620,528 to 962,265 Pa) refrigerant gas pressure for R-12. The capacity of the water regulating valve shall at least equal the water requirement of the condenser, based on available inlet water pressure of 35 psi (241,316 Pa) and a pressure drop across the valve of not greater than 10 psi (68,947 Pa). The sea water control valve shall be located downstream of the condenser and shall prevent entry of sea water into the refrigeration system. The valve body shall be made of valve bronze conforming to MIL-B-16541 and the internal parts of the valve shall be made from nickel-copper conforming to QQ-N-281. The valve shall have either flanged connections conforming to MIL-F-20042 or union ends conforming to MIL-F-1183.

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The strainer body shall be made from 90-10 copper-nickel bronze conforming to ASTM B61, monel or other materials satisfactory to the procuring activity. A strainer of No. 20 mesh nickel-copper screen shall be provided. The strainer basket shall be removable for cleaning without disassembly or connecting piping. The strainer shall be not less than 0.75 inch ips.

3.6.4. Electrical controls. All controls and controllers shall be in accordance with NEMA ICS 1 and ICS 2. The controller enclosure shall be in accordance with type I of NEMA ICS 6. All controls shall be located at the upper portion of the front panel and shall consist of a three position switch and indicating light. The switch may be of the rocker type with three positions "WASH," "OFF," and "AUTO." The indicator light shall illuminate whenever the switch is placed in the "AUTO" position, blending and refrigeration mode.

3.6.5. Temperature control. The temperature control shall thermostatically control the mixing and cooling cycle and maintain the temperature of the liquid product at 40°F (4°C).

3.6.6 Electric motors. Motors shall conform to NEMA MG 1. The blender motor shall be not less than 1.0 horsepower (hp), rubber or shock mounted, ball bearing type and equipped with overload protection. The motor for the refrigeration unit shall be not less than 1.0 hp.

3.6.7 Electrical cords and plugs. Unless otherwise specified (see 3.5.1 and 6.2), the cord shall be 4-wire with ground with an attached plug conforming to NEMA WD 1 configuration L22-30 and shall meet the requirements of UL 498.

3.7 Operation. The blenders and all components shall operate as intended without binding, looseness, or malfunction. Completely assembled blenders and all water connections shall not leak when tested in accordance with 4.6.1 and 4.6.2. The blenders shall blend and cool 20 quarts (18.9 L) of production mix for the type I and 10 quarts (9.5 L) for the type II within 12 minutes. The completely blended product shall have a smooth textured consistency.

3.7.1 Inclined operation. The blender shall operate satisfactorily when inclined at an angle of 15° each side of vertical in each of two vertical planes at right angles to each other, with no spillage of fluid or product.

3.8 Electromagnetic interference suppression. When specified (see 6.2), the blender shall be capable of meeting the UMO5 requirements of MIL-STD-461 for Class C3, Group I equipment.

3.9 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b Current-carrying contact surfaces, such as relay contact points, shall not be coated.

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3.10 Data name plates. Data name plates shall be made of minimum 20 gauge corrosion-resisting metal and attached to each item such a manner as to meet the applicable sanitary requirements for this equipment. The plate shall contain the following information stamped, engraved or applied by photosensitive means:

National Stock Number

Procurement Instrument Identification Number

Manufacturer's Name, Address, Phone Number

Manufacturer's Model Number

Each plate shall be placed so that it is readily visible to the operator during normal operating use. Each plate shall be placed in a manner so as to not adversely affect the life and utility of the item.

3.11 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.12 Workmanship. All components and assemblies of the blender shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, damage, dents, and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes, and spurs. External surfaces shall be free from sharp edges and corners, except when corners are required.

3.12.1 Finish. The blender shall be smooth on all surfaces and edges and shall be free from discoloration or stains. The components shall be finished in accordance with the manufacturer's standard practice.

3.12.2 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.12.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loading.

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3.12.4 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.12.5 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.1.3 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.3. The Government reserves the right to examine and test all blenders to determine the validity of the certification.

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4.1.3.1 NSF. Acceptable evidence of meeting the requirements of NSF C-2 shall be one of the following:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished blender.
b. A certification for the blender issued under their special one-time contract evaluation/certification service.
c. A certified test report, acceptable to the contracting officer with the advice from the Army Surgeon General, indicating that the blenders have been tested and conform to NSF C-2.

4.1.3.2 UL. Acceptable evidence of meeting the applicable requirements of UL shall be the UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the blender has been tested and conforms to UL 621.

4.1.3.3 DHHS. Acceptable evidence of meeting the DHHS requirements shall be a certificate of compliance stating that the plastics, rubber, or rubber like materials used in the blenders comply with the requirements of the Federal Food, Drug, and Cosmetic Act; Food, Additive Amendment for Contact with Food.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete blender when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5.1, and when specified, the preproduction pack inspection of 4.8 (see 4.8 and 6.2). The first article may be either a first production item or a standard production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.2, and the packaging inspection of 4.8. Unless otherwise specified herein (see 4.5.2), this inspection shall be performed on the samples selected in accordance with 4.3.

4.2.2.1 Density of insulation. The density test for insulation shall be performed as specified in 4.7. The inspection lot shall be the amount of material used in one days' production. The sample unit for this test shall be one 6-inch (152 mm) cube of insulation.

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4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete blender. All blenders offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for dimensional examination. Dimensional examination shall be based on inspection level S-2 and an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.3.2 Sampling for visual examination. Visual examination shall be based on inspection level II and an AQL of 2.5 percent defective.

4.3.3 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 4.0 percent defective.

4.4 Examination. Each blender shall be examined for compliance with the requirements in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.1 Dimensional examination. The blender shall be examined for compliance with all dimensional measurements specified in section 3 of this specification. Any dimensions not within the specified tolerance shall constitute one defect.

4.4.2 Visual examination. Each blender shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

4.5 Tests.

4.5.1 First article tests. The first article shall receive the tests of 4.6.1 through 4.6.4, and when specified (see 6.2), the test of 4.6.5. Failure to pass any test shall constitute cause for rejection.

4.5.2 Quality conformance tests. Each blender produced shall receive the tests of 4.6.3 and 4.6.4. Each sample selected in accordance with 4.3 shall receive the tests of 4.6.1, 4.6.1.1, and 4.6.1.2.

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TABLE I. Classification of visual defects

Classification	Defects	Requirement paragraph
MAJOR		
101	Blender not manufacturer's standard commercial product.	3.1
102	Insulation of mixing chamber not foamed in place.	3.4.3 and 3.6.2.1
103	Wiring cut, abraded ,or loose at terminal.	3.5.1
104	Slack not provided for wiring to relieve strain.	3.5.1
105	Component not readily accessible for servicing where required.	3.5.2
106	Part or component missing.	3.6
107	Any bottom flanges missing or not provided with boltholes (unless casters are specified).	3.6
108	Frame not of welded construction.	3.6
109	Casters (when required) not provided with brakes.	3.6
110	Bolts, nuts, screws, studs, and other types of threaded fasteners not locking type nuts or fasteners.	3.6
111	Bolts, nuts, screws, studs, and other types of threaded fasteners missing, broken, stripped, or loose.	3.6, 3.12.4, and 3.12.5
112	Panels not provided.	3.6.1
113	Side panels not provided with louvers.	3.6.1
114	Panel top not provided with filler opening and cover.	3.6.1
115	Filler cover not spring loaded to closed position.	3.6.1
116	Insulation does not completely cover mixing chamber.	3.6.2.1
117	Mixing chamber not fabricated of stainless steel.	3.6.2.1
118	Door assembly not fabricated of clear plastic.	3.6.2.2
119	Cross-bar of door assembly missing.	3.6.2.2
120	Adjusting screw not provided on cross bar of door assembly.	3.6.2.2

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TABLE I. Classification of visual defects - Continued.

Classification	Defects	Requirement paragraph
MAJOR (continued)		
121	Draw valve unit of door assembly not complete (i.e., piston, "O" ring, or handle).	3.6.2.2
122	Door assembly gasket missing.	3.6.2.2
123	Water control not manual type.	3.6.3.4
124	Water control not spring loaded to close position.	3.6.3.4
125	Switch not rocker type.	3.6.4
126	Switch not three position.	3.6.4
127	Indicator light missing.	3.6.4
128	Motors not as specified.	3.6.6
129	Functioning component that does not operate as intended.	3.7
130	Any rough or sharp edges, burrs, slivers or scratches; rough die or grind marks.	3.12
131	Any component fractured, buckled, bent, punctured, malformed, or damaged.	3.12
MINOR		
201	Markings missing, incomplete or not legible.	3.10

4.6 Test methods.

4.6.1 Operational test. Connect the blender to an electrical power source having the same characteristics specified in 3.5.1. Also connect the fresh water and salt water supply lines. Place with switch in the "WASH" position and check to assure that all moving components function as intended. Observe all components for conformance with 3.6. Then place switch in the "AUTO" position and check to assure that all moving components in the refrigeration system function as intended. Operate the blender through one complete cycle, i.e., place switch in "AUTO" position, fill water in mixing chamber to desired level and observe all components for conformance with 3.6. Nonconformance with the requirements specified shall constitute failure of this test.

4.6.1.1 Inclined operational test. After the test in 4.6.1 is completed, position the blender with the base set at an angle of 15°, then operate blender with 75 percent capacity for 30 seconds each at each side of the vertical in each of two vertical planes at right angles to each other. At each of these positions observe for conformance with specified requirements of 3.7.1. Any nonconformance with specified requirements shall constitute failure of this test.

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4.6.1.2 Door leak test. During the tests specified in 4.6.1 and 4.6.1.1 with the blender operating in "AUTO" mode and the mixing chamber filled with liquid observe the area around the plastic door for compliance with the leak requirements specified in 3.6.2.2. Any nonconformance with the leak requirements of 3.6.2.2 shall constitute failure of this test.

4.6.2 Performance test. The blender shall be tested to determine that the machine mixes, blends, cools powdered dehydrated mix product and water at the temperature and production rate specified in 3.5. The blender shall be at equilibrium in an ambient temperature of not less than 75°F (24°C) before the start of the test. An ambient temperature of not less than 75°F (24°C) shall be maintained during the test. The blender shall be started and the water fill button depressed until water reaches the full indicator line in the mixing chamber door. After the proper water level is attained place switch in "AUTO" position starting the blending action and the refrigeration system. Through the funnel located on the top of the blender, fill the blender with the required amount of dehydrated powdered mix to produce 20 quarts (18.9 L) for the type I and 10 quarts (9.5 L) for the type II. Failure of the blender to blend and cool the batch to a smooth texture to 40°F (4°C) within 12 minutes or failure to comply with any requirements specified herein shall constitute failure of this test.

4.6.3 Refrigerant leak test. Each blender shall be refrigerant leak tested to determine conformance with 3.6. The test for leakage shall be made with a halide leak detector or an electronic leak detector or any other method standard with the manufacturer. Any leakage shall constitute failure of this test.

4.6.4 Refrigerant moisture test. The refrigerant moisture and liquid indicator shall be checked for dryness to determine conformance with 3.6.3.3 as shown by the manufacturer's color chart data. A reading indicating moisture shall constitute failure of this test.

4.6.5 Electromagnetic interference suppression. When electromagnetic interference suppression is required (see 3.8), the blender shall be tested to meet the requirements of 3.8 in accordance with MIL-STD-462.

4.7 Density test for insulation. Prior to each day's production, a cube of insulation not less than 8 inches (203 mm) shall be taken. From the center of the sample a 6-inch (152 mm) cube shall be cut and weighed. This weight shall be multiplied by 8 to determine conformance with the density requirement of 3.4.3.1. In addition, a daily record of the foam insulation shall be kept as follows and the production unit shall not deviate more than plus or minus 5 percent from the date recorded.

a. Calibration of mix (mix proportions)
b. Amount of flow (per box)
c. Time of flow (per box)
d. Mould temperature (per box)

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Inability of sample to conform with density requirement of 3.4.3.1, or more than 5 percent deviation of data gathered on the production unit shall constitute failure of this test.

4.8 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-R-12323. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with the requirements of MIL-R-12323 with the level of preservation and packaging and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The blender is intended for use in galleys to blend and cool water and powdered dehydrated mix products.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and number of blenders required (see 1.2).
c. Electrical characteristics, if other than specified (see 3.5.1).
d. When the type I blender is to be furnished with casters (see 3.6).
e. When the dimensions of the blenders are other than as specified (see 3.6).
f. When electrical cords and plugs are to be other than as specified (see 3.6.7).
g. When electromagnetic interference suppression and testing is required (see 3.8 and 4.5.1).
h. When fungus and moisture resistance is required (see 3.9).
i. When a first article sample and first article inspection is required. When a preproduction pack inspection is required and the time frame required (see 4.2.1 and 4.8).
j. Level of preservation and packaging and the level of packing (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), and invokes the provisions of DOD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DOD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

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6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Subject term (key word) listing.

Appliance, commercial
Blender, sea water cooled
Food service equipment

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD

Review Activities:
Navy - MC, SA, SH

Preparing Activity:
Navy - YD

(Project 7310-0707)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44037A	BLENDER, POWDERED MIX, MECHANICALLY REFRIGERATED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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